Degree reduction and recursive parity queries for explicit Boolean polynomials

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Abstract

We give an algorithmic corollary of structural results of Zhang and Moshkovitz–Zhu. Given the explicit squarefree algebraic normal form of a degree-d polynomial p on $\mathbb{F}_2^n$, with $d \geq 3$, a randomized algorithm constructs a binary subspace on every coset of which the degree drops. For $f = (-1)^p$, Fourier spectral norm B, and error parameter δ, the codimension is at most $\min\{n, 2^{2d-3}\lceil \log_2(4n(d-2)/\delta)\rceil \log_2 B\}$. The algorithm runs in time polynomial in n, the explicit input length and $\log(1/\delta)$, and succeeds with probability at least $1-\delta$. It samples a contraction over a finite extension field, computes its kernel and intersects that kernel with binary space. A maximal-minor argument preserves the imported vanishing identity; determinant evaluation in characteristic two avoids exponential finite-difference enumeration. We also construct an on-demand recursive parity policy from the original ANF and a compact affine map. For every fixed hidden input, with probability at least $1-\delta$, it evaluates p correctly using $\min\{n, O(1+4^d \log s \log(nd))\}$ parity queries, where s is Fourier sparsity. Its bit time is $2^{O(d\log(d+1))}\,\mathrm{poly}(n, S_0, \log(1/\delta))$ and its working space is polynomial, with an explicit precision interface. Boolean finite differences followed by scalar extension avoid constructing residual ANFs. Bounded repetition carries accuracy without enlarging the field. The query cap n holds unconditionally; the smaller bound and correctness are a joint per-input guarantee, not a globally correct sampled tree. The stronger existential query bound and unrestricted XOR log-rank remain outside the result.

1 Statement and relationship to prior work

All logarithms are base two unless written ln. For $f : \mathbb{F}_2^n \to \{-1, 1\}$ let

$$\widehat{f}(\xi) = 2^{-n} \sum_x f(x)(-1)^{\xi \cdot x}, \qquad B = \|\widehat{f}\|_1, \qquad s = |\operatorname{supp} \widehat{f}|.$$

A parity query asks for a binary linear form in the unknown input. Independent equations for a subspace of codimension r specify r such queries; their answers identify a coset. Degree throughout means the degree of the unique squarefree algebraic normal form (ANF) over $\mathbb{F}_2$, not real Fourier degree.

The degree-reduction approach to parity decision trees predates this note [3]. Zhang [1] proves an existence bound of order $1+4^d \log s$, using a generic contraction radical and dimension-preserving descent. Remark 3.17 of that work expressly leaves construction time open. We replace the descent step by a directly computable binary intersection, accepting a logarithmic loss. The generic-rank comparison used below is due to Moshkovitz and Zhu [2]. The contribution here is explicit input-access, specialization and resource analysis, first for one batch and then for a complete visited path. The latter computes restricted top forms from Boolean evaluations of the original source. No priority claim for the combination is made.

Theorem 1. *Let $p \in \mathbb{F}_2[x_1, \ldots, x_n]$ be given by its L distinct squarefree nonzero monomials, of actual degree $3 \le d \le n$. Let $0 < \delta \le 1/4$ be rational and let S be the encoding length, including δ. Put*

$$k = \left\lceil \log \frac{4n(d-2)}{\delta} \right\rceil.$$

There is a randomized algorithm whose worst-case bit cost is polynomial in n, S and $\log(1/\delta)$. It returns a basis matrix C for a binary subspace U and independent equations $Hx = 0$ for U, or declares failure. With probability at least $1 - \delta$ it returns matrices satisfying, simultaneously for every $a \in \mathbb{F}_2^n$,

$$\deg_{\mathbb{F}_2}(p(a + Cz)) \le d - 1, \tag{1}$$

$$r := \operatorname{rank}_{\mathbb{F}_2} H = \operatorname{codim} U \le \min\{n, k\, 2^{2d-3} \log B\} \le \min\{n, k\, 2^{2d-4} \log s\}. \tag{2}$$

The degree guarantee is exact on success. Both declared failure and incorrect output count against δ. Space is polynomial and the output length is $O(n^2)$ bits. The algorithm needs no Fourier data, B, s, or generic-rank oracle.

The input model is essential. Polynomial time includes polynomial dependence on the ambient dimension n; S alone need not describe all unused variables. This is an explicit ANF input, not an arbitrary succinct arithmetic or Boolean circuit. The codimension estimate remains exponentially dependent on d, even though the construction time does not.

For recursion, separate the original polynomial's encoding length S_0 from the accuracy input. An integer $b \ge 2$ requests failure at most $\epsilon = 2^{-b}$. For any desired $0 < \delta < 1$, choose $b = \max\{2, \lceil \log(1/\delta) \rceil\}$, so $\epsilon \le \delta$. A binary-fraction prefix through the first nonzero bit also suffices. An arbitrary numerator/denominator encoding of length A, if used instead, adds its own $\mathrm{poly}(A)$ parsing cost; it cannot be charged solely to $\log(1/\delta)$.

Theorem 2. *Given the original explicit ANF of degree $3 \le d \le n$, precision $b \ge 2$, and parity-query access to a hidden $x \in \mathbb{F}_2^n$, there is a randomized on-demand policy with the following properties. Set*

$$k = \lceil \log(8nd) \rceil, \qquad t_F = 2k(b+1), \qquad t_A = b + \lceil \log(2d) \rceil.$$

For every fixed x, with probability at least $1 - 2^{-b}$, the policy returns $p(x)$ correctly and uses at most

$$\min\left\{n,\ 1 + \left[1 + \frac{k(2^{2d-1} - 8)}{3}\right] \log B\right\} \le \min\left\{n,\ 1 + \frac{1}{2}\left[1 + \frac{k(2^{2d-1} - 8)}{3}\right] \log s\right\} \tag{3}$$

queries, hence $\min\{n, O(1 + 4^d \log s \log(nd))\}$. On every random tape it terminates, asks at most n queries, uses polynomial working space, and has bit time

$$2^{O(d \log(d+1))}\, \mathrm{poly}(n, S_0, b),$$

with a polynomial exponent independent of d. It uses neither Fourier information nor a residual-degree oracle and prints only the visited path. Original degree at most two is handled exactly in polynomial time by the leaf routine below.

Theorems 1 and 2 have different computational guarantees: the one-batch construction is uniformly polynomial, whereas recursion pays an explicit original-degree factor. Neither claims that a single random seed supplies a correct deterministic tree for all inputs.

2 The imported structural interface

Define $\Delta_h p(x) = p(x+h) + p(x)$ and

$$T(h_1, \dots, h_d) = \Delta_{h_1} \cdots \Delta_{h_d} p(0).$$

Formal evaluation uses the supplied squarefree polynomial over any extension of $\mathbb{F}_2$. The form T is symmetric, multilinear and alternating. Let $h = d-2$, let $y_1, \dots, y_h$ be independent formal n-vectors, and put

$$K = \mathbb{F}_2(y_1, \dots, y_h), \quad N(y)_{ij} = T(y_1, \dots, y_h, e_i, e_j), \quad R = \operatorname{rank}_K N, \quad W = \ker_K N.$$

The needed external inputs are

$$T(K^n, W, \dots, W) = 0, \qquad R \le 2^{d-2} M, \tag{4}$$

where the identity has $d-1$ copies of W, and M is the maximum rank of N at binary anchors. The first statement is Zhang's Theorem 3.12 [1]. The second is the generic/max-rank inequality of Moshkovitz–Zhu [2], applied to $h = d-2$ homogeneous multilinear blocks over $\mathbb{F}_2$. Its factor is 2^h and it has no large-field restriction. We import these structural statements rather than present independent proofs of them.

For completeness, the remaining quantitative connection is elementary. At binary anchors, $g = \Delta^{\times}_{y_1} \cdots \Delta^{\times}_{y_h} f$ is a quadratic phase, where $\Delta^{\times}_u f(x) = f(x+u)f(x)$. Its polar matrix is $N(y)$. If the polar rank is ρ, quadratic normal form gives $\|\widehat{g}\|_1 = 2^{\rho/2}$: each independent quadratic pair contributes a factor of two and affine terms only translate the spectrum. Translation preserves spectral norm and multiplication is Fourier convolution. Hence

$$2^{\rho/2} \le B^{2^h}, \qquad M \le 2^{d-1} \log B, \qquad R \le 2^{2d-3} \log B.$$

Parseval and Cauchy–Schwarz give $B \le \sqrt{s}$. These inequalities are used in the analysis; their parameters are never computed by the algorithm.

3 Specialization and binary equations

Lemma 3. *Let $N(y)$ be a polynomial matrix over $\mathbb{F}_2$ of generic rank R. Suppose its generic kernel W satisfies the first identity in (4) for a fixed multilinear form T over $\mathbb{F}_2$. For any field extension E and any specialization a with $\operatorname{rank}_E N(a) = R$, the kernel $W_a = \ker_E N(a)$ satisfies $T(E^n, W_a, \dots, W_a) = 0$.*

Proof. If $R = 0$, the matrix is identically zero and the assumed identity gives the claim. Otherwise choose R rows and columns with submatrix P such that $D = \det P$ is nonzero at a. Reorder columns and write those rows as $[P\ Q]$. Over the localization $\mathbb{F}_2[y, 1/D]$, the columns of

$$V = \begin{bmatrix} -P^{-1}Q \\ I_{n-R} \end{bmatrix}$$

are a basis of the generic kernel. They solve the selected equations, whose rows span the generic row space, so $NV = 0$. Every entry has denominator dividing D. For standard basis vectors e_i and columns of V, multiply each identity $T(e_i, V_{j_1}, \dots, V_{j_{d-1}}) = 0$ by D^{d-1}. It becomes a polynomial identity, which can be specialized at a. Since $D(a) \neq 0$, the original identity also holds there. The $n - R$ independent columns of $V(a)$ solve $N(a)V(a) = 0$ and span its entire kernel. Multilinearity proves the result. If $R = n$, the kernel has dimension zero and the assertion is immediate. □

This establishes preservation at every full-generic-rank point. No further denominator condition or union over pivot charts is needed. For probability analysis only, fix any nonzero generic R-minor D of N (use $D = 1$ if $R = 0$). Its degree is at most $R(d-2) \le n(d-2)$. For independent uniform anchors in $E = \mathbb{F}_{2^k}$, the polynomial zero bound gives

$$\Pr[\operatorname{rank} N(a) < R] \le \Pr[D(a) = 0] \le \frac{n(d-2)}{2^k} \le \delta/4. \tag{5}$$

The zero bound is the usual induction from the univariate root bound. The algorithm does not find D, determine R, or test this event. All larger minors vanish identically, so no specialization has rank greater than R.

For $A = N(a)$, select a row basis over E, of size r_E. Expand its entries in a fixed binary basis of E. For binary vectors x, each E-linear equation is exactly equivalent to at most k binary equations. Binary elimination produces H and C with

$$U = \ker_{\mathbb{F}_2} H = \ker_E A \cap \mathbb{F}_2^n, \qquad \operatorname{codim} U \le \min\{n, kr_E\} \le \min\{n, kR\}.$$

Thus (2) holds on every nonfailure output, even on exceptional rank samples. On the full-rank event Lemma 3 gives $T(U^d) = 0$. To obtain (1), each degree-d coefficient of the Boolean restriction $p(a + Cz)$ is a d-fold difference in the corresponding distinct columns of C. It equals T on those columns, independently of a, and therefore vanishes. The restriction has degree at most d before this argument. If $\dim U < d$, no such coefficient exists.

4 Polynomial evaluation and a capped algorithm

Lemma 4. *If $p = \sum_I \prod_{i \in I} x_i$ is squarefree of degree d, then over any field of characteristic two,*

$$T(h_1, \dots, h_d) = \sum_{|I|=d} \det\big[(h_j)_{i_\ell}\big]_{j,\ell=1}^{d}, \qquad I = \{i_1, \dots, i_d\}. \tag{6}$$

Consequently one value of T can be computed in $O(Ld^3)$ field operations.

Proof. Expand the subset formula for the d differences of one monomial. Every term omitting a direction occurs an even number of times and cancels. A monomial of degree less than d contributes nothing. For a degree-d monomial, each surviving term uses every direction exactly once, so terms are indexed by permutations. This is the permanent of the displayed matrix, equal to its determinant in characteristic two. Summing over monomials and using Gaussian elimination proves the cost. □

Only the listed degree-d monomials are needed. In particular, neither 2^d subset enumeration nor construction of an n^d tensor is necessary. Let

$$b = \lceil \log(2/\delta) \rceil, \qquad t = 2kb.$$

The following algorithm has fixed caps; all arithmetic is exact.

1. Read and check the explicit squarefree ANF and its actual degree. Compute k, b, t using integer comparisons with the rational input.

2. Try at most t independently uniform monic degree-k binary polynomials g, using k fair bits per trial. Test irreducibility by checking $\gcd(g, X^{2^j} - X) = 1$ for $1 \le j \le \lfloor k/2 \rfloor$, using repeated squaring modulo g. Keep the first passing polynomial; if none passes, declare failure.

3. Work in $E = \mathbb{F}_2[X]/(g)$ with its polynomial basis. Draw $d-2$ uniform independent anchors in E^n, using fresh random bits.

4. Form $A_{ij} = T(a_1, \ldots, a_{d-2}, e_i, e_j)$ by (6). Compute a row basis over E, expand it into binary equations, and compute H and C by binary elimination. Return these matrices without retrying on the basis of sampled rank.

The field test is exact: a reducible degree-k polynomial has an irreducible factor of degree $j \le k/2$, which divides $X^{2^j} - X$. An irreducible degree-k polynomial has none. Our parameters ensure $k \ge 6$. At least a $1/(2k)$ fraction of monic degree-k binary polynomials are irreducible. To see this, in $\mathbb{F}_{2^k}$ the elements of smaller degree lie in proper subfields. Their number is at most $\sum_{j=1}^{\lfloor k/2 \rfloor} 2^j \le 2^{k-1}$. Each irreducible degree-k polynomial accounts for k of the remaining elements. This is a counting argument, not field enumeration. Failure of all t trials has probability at most $\exp(-t/(2k)) = \exp(-b) \le 2^{-b} \le \delta/2$. Conditional on any verified field, fresh anchors satisfy (5). The total bad probability is at most $3\delta/4 \le \delta$.

Proposition 5. *Besides polynomial input-reading and validation overhead, the algorithm uses at most*

$$O\Big(tk^4 + (n^2Ld^3 + n^3)k^3 + kn^3\Big) \tag{7}$$

arithmetic bit operations with indexed storage. It uses at most $tk + (d-2)nk$ random bits and polynomial working space. Ordinary sequential access can also be charged while preserving a polynomial bit-time bound.

Proof. Schoolbook repeated squaring and polynomial gcd give the deliberately loose $O(k^4)$ bound per field trial. Field addition and multiplication cost at most $O(k^2)$ bits; inversion by exponentiation or polynomial Euclid fits in $O(k^3)$. Lemma 4 computes n^2 entries in $O(n^2Ld^3)$ field operations. Row elimination costs $O(n^3)$ such operations. Expanding rows gives at most kn binary equations of length n; elimination and a nullspace basis cost $O(kn^3)$ bit operations. The trial and anchor bit counts are explicit. Store one trial polynomial, all anchors, the two elimination matrices and one determinant workspace. A conservative space bound is $O(S + n^2k + ndk + d^2k + k^2 + b)$ bits. Address decoding or a sequential scan of these polynomial-size arrays adds only polynomial overhead. Since $d \le n$, L is bounded by the explicit description length and $k, b = O(\log(nd/\delta))$, these are polynomial costs in the parameters stated in Theorem 1. □

This proves Theorem 1. It supplies an algorithm and resource proof, not an implementation or performance benchmark.

5 Boolean evaluation before scalar extension

Maintain $a \in \mathbb{F}_2^n$ and a full-column-rank matrix $C \in \mathbb{F}_2^{n \times m}$ such that the possible hidden inputs are exactly $a + C\mathbb{F}_2^m$. The current Boolean function is $u(z) = p(a + Cz)$. Its unique squarefree representative is used in the proof but never expanded or stored. Evaluating p at a fully specified Boolean point is charged computation from its explicit description, not a query to the hidden x.

Lemma 6. *Suppose the Boolean degree of u is at most r. For binary directions define*

$$F_r(v_1, \ldots, v_r) = \sum_{J \subseteq [r]} p\left(a + C\sum_{j \in J} v_j\right) \quad \text{in } \mathbb{F}_2. \tag{8}$$

This is the canonical top r-form: it is independent of the base point, symmetric, alternating and $\mathbb{F}_2$-multilinear. It is zero if $\deg u < r$. *For a binary basis $\beta_0, \dots, \beta_{k-1}$ of E, decompose $h_i = \sum_j \beta_j v_{i,j}$. Its scalar extension is exactly*

$$T_E(h_1, \dots, h_r) = \sum_{j_1,\dots,j_r=0}^{k-1} \left(\prod_{i=1}^{r} \beta_{j_i} \right) F_r(v_{1,j_1}, \dots, v_{r,j_r}). \tag{9}$$

For $T_E(h_1, \dots, h_{r-2}, e_i, e_j)$, at most $k^{r-2}2^r$ original-source evaluations suffice. They can be streamed in polynomial space.

Proof. An $(r+1)$-fold difference of a Boolean polynomial of degree at most r vanishes. Thus the r-fold difference is independent of its base point. The operator identity $\Delta_{v+w} = \Delta_v + \Delta_w + \Delta_v\Delta_w$ proves additivity in every direction: its last term contributes an $(r+1)$-fold difference. Also $\Delta_v^2 = 0$, proving alternation; commutation gives symmetry. Values on distinct basis vectors extract the top coefficients of the canonical ANF. Formula (9) is now multilinearity, and in the contraction only the first $r-2$ slots need expansion. Nested tuple/subset loops accumulate one field value without storing their exponentially many states. □

This order of operations matters. For $p = x_1x_2 + x_1$ and $(x_1, x_2) = (z, z)$, the Boolean restriction and every value of (8) are zero. The unreduced formal expression $z^2 + z$ is nonzero at $\alpha \in E \setminus \mathbb{F}_2$. More sharply, restrict $x_1x_2x_3x_4$ to (z_1, z_1, z_2, z_3). Its canonical cubic form at $(\alpha e_1, e_2, e_3)$ equals α, whereas third field differences of the unreduced expression return α^2. Lemma 6 handles both controls in every dimension; it never evaluates the substituted formal expression at a nonbinary point.

An upper bound on the current degree suffices. If the actual degree is smaller, the scheduled form is zero. On a history where the upper bound fails, the same finite loops still terminate; their output need not be the canonical form, and such histories are included in the failure budget.

6 The recursive policy and its correctness

Use the parameters of Theorem 2 throughout this section. Construct one verified field by the earlier irreducibility test, now with cap t_F. The counting argument gives

$$\Pr[\text{field failure}] \le e^{-(b+1)} \le 2^{-(b+1)} = \epsilon/2.$$

On failure output zero with a failure flag. Conditional on any verified field, consider a stage with current degree at most r. If the degree is exactly r, the structural interface and Lemma 3 apply in dimension m; if smaller, its form is zero and generic rank $R = 0$. One nonzero generic minor has degree at most $m(r-2) \le nd$. A uniform anchor sample therefore misses generic rank with probability at most $nd/2^k \le 1/8$.

Compute exactly t_A independent candidate contractions by Lemma 6, and keep an exact row basis from a candidate of largest observed rank, with deterministic tie breaking. Every sampled rank is at most R, since all larger minors vanish identically. If any sample attains R, the selected one does too. Hence

$$\Pr[\text{selected rank} < R] \le 8^{-t_A} \le 2^{-t_A} \le \epsilon/(2d). \tag{10}$$

This is probabilistic amplification, not a certificate of generic rank. No unbounded retry is used. The field degree is independent of b; choosing it proportional to $\log(nd/\delta)$ inside (9) would instead introduce a degree-dependent power of the precision parameter.

6.1 Lifting queries and updating the affine map

Binary elimination constructs L_C with $L_C C = I_m$. To ask for λz, ask the allowed original-input query $(\lambda L_C)x$ and add the known offset $(\lambda L_C)a$. Expand the selected field row basis into binary equations and row-reduce to independent rows H, of rank ρ. The answers determine $Hz = y$. Compute a solution z_0 and a nullspace basis V, then replace

$$a \leftarrow a + Cz_0, \qquad C \leftarrow CV, \qquad m \leftarrow m - \rho. \tag{11}$$

All answer vectors are consistent because H has full row rank. The new columns remain independent and describe exactly the remaining input set. The map uses at most $n + n^2$ bits, independent of the number of previous batches. Even on a bad rank history, the true x remains in this set.

On a good stage, binary intersection and the vanishing identity lower the degree on every coset to at most $r - 1$. The same original source and the updated map therefore supply the next form through Lemma 6. After two stages the map is $a + Cz_0 + CVz_1$, with linear part CVV_1 and degree at most $r-2$. This proves arbitrary repeated composition without expanded ANFs or actual-degree detection.

6.2 Algorithm and conditional history

1. Normalize the original ANF, determine its degree d, and retain it throughout. If $d \leq 2$, use the exact leaf routine below.

2. Construct the verified field under cap t_F. On success set $a = 0$, $C = I_n$ and $m = n$.

3. For scheduled $r = d, d-1, \ldots, 3$: if $m = 0$, return $p(a)$. Otherwise compute t_A independent candidate matrices by (8)–(9); retain a maximum-rank row basis; expand and reduce its binary equations; ask the lifted parities; and apply (11).

4. Apply the quadratic/affine leaf routine to the current source evaluator and return its bit.

Fix the hidden x. Condition on successful field construction and on the complete history of earlier successful stages. The current field, a, C, u are fixed, the degree promise follows inductively, and future anchor bits remain independent and uniform. Equation (10) applies to this conditional history. A first-failure decomposition over at most $d - 2$ visited stages yields

$$\Pr[\text{field failure or a visited bad stage}] \leq \epsilon/2 + (d-2)\epsilon/(2d) \leq \epsilon. \tag{12}$$

On its complement the final degree is at most two, so the leaf is exact. On other histories all loops, eliminations and affine updates still terminate. Only correctness, the propagated degree promise and the smaller query estimate may fail. This proof concerns every fixed input separately. It does not union-bound over an entire tree or supply an all-input correct random seed.

6.3 Exact quadratic and affine leaves

For $m = 0$ return $p(a)$. Otherwise evaluate u at zero, the e_i and the $e_i + e_j$ for $i < j$, and set

$$c = u(0), \quad l_i = u(e_i) + c, \quad b_{ij} = u(e_i + e_j) + u(e_i) + u(e_j) + c.$$

The explicit quadratic $g(z) = c + \sum_i l_i z_i + \sum_{i<j} b_{ij} z_i z_j$ equals u on a good history. On a bad history it is simply a defined quadratic approximation. Form its alternating polar matrix B_g. Symplectic elimination constructs invertible P with $P^{\mathsf{T}} B_g P$ consisting of t alternating pairs and a zero block,

where $\operatorname{rank} B_g = 2t$. Explicitly choose a pair a_0, b_0 with pairing one, and replace each remaining vector v by

$$v + B_g(v, b_0)a_0 + B_g(v, a_0)b_0.$$

It is then perpendicular to that pair. Repeating preserves a basis and isolates the radical; explicit bookkeeping costs at most $O(m^4)$ bit work. The canonical Boolean expression is

$$g(Pw) = c + \sum_{i=1}^{t} w_{2i-1}w_{2i} + \ell \cdot w, \qquad \ell_j = g(Pe_j) + c.$$

Computing these linear terms includes squares reducing to linear terms. Query the t odd coordinates, lifting through CP. Once they are known, the remaining expression is affine: at most one additional parity gives its value. The leaf uses at most $t+1$ queries and at most m; if $t = 0$ it uses zero queries for a constant and at most one otherwise.

On a good history, $\|\widehat{(-1)^u}\|_1 = 2^t$. Indeed, each independent two-bit quadratic pair has four Fourier coefficients of magnitude $1/2$ and norm two; affine characters shift frequencies and invertible linear changes permute them. Affine restriction cannot increase spectral norm: it groups original Fourier coefficients with signs, so the triangle inequality applies. Thus the leaf costs at most $1+\log B$ with B from the original input.

6.4 Total queries

On the good event, stage r uses at most $k2^{2r-3}\log B$ queries, including zero when the actual degree is smaller than r. Summing with the leaf gives

$$1 + \left[1 + k\sum_{r=3}^{d} 2^{2r-3}\right]\log B = 1 + \left[1 + \frac{k(2^{2d-1} - 8)}{3}\right]\log B.$$

Every batch removes exactly as many dimensions as queries asked, and the leaf uses at most the remaining dimension. Hence the cap n holds on all tapes. Taking the minimum and using $B \le \sqrt{s}$ proves (3). The additive one covers a nonconstant parity with $s = 1$.

7 Total recursive computation and space

Here S_0 is the original polynomial encoding length. A normalization pass can check indices, sort within monomials and cancel duplicate monomials by in-place scans in polynomial time and $O(S_0)$ storage; no $L \times n$ incidence table is required. Put

$$V_0 = n^2 + (d+1)(S_0 + n) + 1.$$

With indexed bit storage, $O(V_0)$ work covers one original-source evaluation, its binary affine transformation and a subset-sum point for $r \le d$. One contraction entry costs at most

$$O\big(k^{r-2}(2^r V_0 + rk^2)\big) \le O\big((2k)^r(V_0 + dk^2)\big).$$

This counts each binary evaluation, each field-basis product and accumulation. There are at most n^2 entries per sample, dt_A samples, and $O(n^3k^3)$ bit work for each field elimination. Binary expansion/elimination costs $O(kn^3)$ per stage; left inverses, query lifting, coset solutions and affine composition each fit $O(n^3)$. Quadratic coefficient extraction and the leaf cost $O(n^2V_0 + n^4)$. All

query descriptions together use at most n^2 bits, plus n answer bits. Before sequential-access/control overhead a conservative bound is

$$T_0 = \operatorname{poly}(n, S_0, b) + O(t_F k^4 + dt_A[n^2(2k)^d(V_0 + dk^2) + n^3k^3] + dkn^3 + n^2V_0 + n^4). \tag{13}$$

The random-bit cap is

$$t_F k + t_A kn(d-2)(d-1)/2.$$

Keep the original source, current map, one candidate matrix, one best row basis, anchors, binary workspaces, field polynomial and scalar accumulators. Tuples/subsets and trials are streamed. A conservative live-space bound, including transient old/new maps and bounded counters, is

$$W = O(S_0 + kn^2 + ndk + k^2 + b + n^2 + d\log(k+1) + 1). \tag{14}$$

This stores neither a full restricted ANF nor the random tape nor the tree. Sequential scans, index decoding and bounded-width control add at most a polynomial overhead; $W^2(T_0 + 1)$ is a conservative ordinary bit-Turing bound.

Finally, $k \le \log(16nd)$ and

$$(2k)^d \le \left(\frac{2}{\ln 2}\right)^d [\ln(16nd)]^d \le 16nd\left(\frac{2d}{e\ln 2}\right)^d.$$

The last inequality follows by maximizing $y^d e^{-y}$ over $y \ge 0$. Since $t_A = O(b+\log d)$, $t_F = O(k(b+1))$ and $d \le n$, equations (13)–(14) prove the time and space claims of Theorem 2 with fixed polynomial exponents. The bounds hold on bad tapes as well. There is no degree or variance loss and no numerical approximation; all errors arise from the capped field search and rank specializations.

8 Controls and limitations

Write $x_I = \prod_{i\in I} x_i$. Four exact controls are:

Polynomial	Anchors	Generic rank	Binary kernel
x_{123}	e_1	2	$\langle e_1\rangle$
$x_{123} + x_{456}$	$e_1 + e_4$	4	$\langle e_1, e_4\rangle$
$x_{123} + x_{145}$	e_1	4	$\langle e_1\rangle$
$x_{1234} + x_{1256}$	e_1, e_2	4	$\langle e_1, e_2\rangle$

The contractions attain these ranks by nonsingular alternating pairs. Matching generic upper bounds follow from alternating rank in dimensions three and five, the two independent three-dimensional blocks in the second example, and the two independent generic anchors in the radical in the quartic example. The listed kernels lower degree on every coset. Adding terms of degree at most $d-1$ leaves T unchanged and preserves the target. In the disjoint-cubic example, contraction only at e_1 has rank two and leaves the second cubic intact: nonzero sampled rank is insufficient.

The field-to-binary loss is real. In the first cubic, take an anchor $(1, \alpha, 0)$ with $\alpha \in E \setminus \mathbb{F}_2$. The extension-field rank is two and the kernel is $E(1, \alpha, 0)$, but its binary intersection is zero. Binary codimension is three, not two. This does not establish sharpness of the factor k. Zero anchors give rank zero and can fail degree reduction; these are covered by the exceptional event.

The estimate is nonvacuous when $4^d \log s \log(nd/\delta) = o(n)$. This regime need not arise by adjoining unused variables: the product of three parities on disjoint growing variable blocks depends

on every variable, has degree three, sparsity eight and at most n^3 ANF monomials. The input cost remains paid.

For an explicit two-stage control take

$$p = x_1x_2(x_3x_4 + x_5x_6) + x_7x_8x_9.$$

Quartic anchors e_1, e_2 give rank four, with pairs $(3, 4)$ and $(5, 6)$. This is generic rank: the two generic anchors lie in the radical of the six-dimensional support, and the last three coordinates are inactive. Querying x_3, x_4, x_5, x_6 leaves

$$u(z) = cz_1z_2 + z_3z_4z_5, \qquad c = x_3x_4 + x_5x_6.$$

The cubic contraction at e_3 has generic rank two, since its alternating three-dimensional block has rank at most two. Querying z_4, z_5, equivalently x_8, x_9, leaves $cw_1w_2 + (x_8x_9)w_3$, of degree at most two for every answer branch. This control illustrates the general composition proof; finite agreement is not used as its substitute.

The recursive construction stores a compact affine map, not a compact residual ANF. Its stronger runtime dependence in d is retained. The known existential query bound is better by a factor $O(\log(nd))$; dimension-preserving descent and the construction of a globally correct deterministic tree remain different obligations. Reading all n hidden bits already permits direct evaluation of the explicit p. Our result concerns fewer parity queries with paid computation, not faster unrestricted evaluation or a general quantum simulator.

Earlier parity-tree recursion is due to Tsang–Wong–Xie–Zhang [3]. Zhang's quantum protocol [4] assumes an efficiently encoded/decoded Fourier spectrum, a different access model and output. Hrubeš's multilinearization hardness result [5], Theorem 7(ii), concerns uniformly small arithmetic representations of whole multilinearizations. It does not rule out our degree-parameterized, promise-based top-form evaluator. These comparisons explain the input distinction; they do not establish novelty. Unrestricted XOR log-rank, stronger AA bounds and BPP versus BQP are not resolved.

Disclosure and status. OpenAI Codex assisted with derivation, source comparison, drafting and document preparation. Internal analytic checks cover the mathematics, sources, claims and document package. No implementation, benchmark, independent peer review or formal certification is claimed. This preprint presents attributed algorithmic corollaries; priority and publication significance are not established.

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